

```
import java.util.*;

public class ContactManager {

    public static void main(String[] args) {

        HashMap<String, Contact> contacts = new HashMap<>();

        // Step 4: Add contacts
        contacts.put("Ada Lovelace", new Contact("Ada Lovelace", "+1 617 555 0101"));
        contacts.put("Alan Turing", new Contact("Alan Turing", "+1 206 555 0102"));
        contacts.put("Grace Hopper", new Contact("Grace Hopper", "+1 703 555 0103"));
        contacts.put("Linus Torvalds", new Contact("Linus Torvalds", "+358 9 555 0104"));
        contacts.put("Bjarne Stroustrup", new Contact("Bjarne Stroustrup", "+45 45 555 0105"));

        // Step 5: Look up a contact
        System.out.println("=== Contact Lookup ===\n");

        // Lookup existing contact
        String searchName = "Grace Hopper";
        Contact found = contacts.get(searchName);
        if (found != null) {
            System.out.println("Found: " + found);
        } else {
            System.out.println("Contact not found.");
        }
    }
}
```

```
}
```

```
System.out.println();
```

```
// Lookup non-existing contact
```

```
String notFound = "Steve Jobs";
```

```
Contact result = contacts.get(notFound);
```

```
if (result != null) {
```

```
    System.out.println("Found: " + result);
```

```
} else {
```

```
    System.out.println("Looking for: " + notFound);
```

```
    System.out.println("Contact not found.");
```

```
}
```

```
// Step 6: Print sorted list
```

```
System.out.println("\n=== All Contacts (Alphabetical Order) ===\n");
```

```
ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
```

```
sorted.sort((a, b) -> a.getName().compareTo(b.getName()));
```

```
for (Contact contact : sorted) {
```

```
    System.out.println(contact);
```

```
}
```

```
}
```

```
}
```

```
}
```